

The screenshot displays the SQL Server Enterprise Manager interface. The central pane shows a query window with the following SQL code:

```
SELECT *
FROM FactResellerSales
WHERE CarrierTrackingNumber = '4911-403C-98'

CREATE NONCLUSTERED INDEX idx_FactReseller_CTA
ON FactResellerSales (CarrierTrackingNumber)
```

Below the query window, the 'Execution plan' tab is active, showing a query plan for 'Query 1: Query cost (relative to the batch): 100%'. The plan consists of a single step: 'SELECT' with a cost of 0%. The 'Index Seek' operator is highlighted, showing it reads 12 of 16 rows (75%) from the 'FactResellerSales' table. The 'Key Lookup' operator follows, with a cost of 94%, reading 12 of 16 rows (75%) from the 'FactResellerSales' table. The 'Properties' pane on the right shows connection details for 'DESKTOP-8488'.

Now we have index seek in above

Types of Scan

-  **Table Scan:** Reads every row in a table.
-  **Index Scan:** Reads all entries in an index to find results.
-  **Index Seek:** Quickly locates specific rows in an index.

So, there are three types of scan. The best one is index seek and the worst one is table scan.

In index scan, the number of rows to be executed is 12. We are reading 12 rows only in order to get 12.

Cost is very less and it's very fast.

Cpu cost for index -

SQLQuery2.sql - D:\BQBU\Youtube (52)* SQLQuery1.sql - D:\BQBU\Youtube (58)*

```
SELECT *
FROM FactResellerSales
WHERE CarrierTrackingNumber = '4911-403C-9...
```

CREATE NONCLUSTERED INDEX idx_FactReseller
ON FactResellerSales (CarrierTrackingNumber)

Query 1: Query cost (relative to the batch): 1

SELECT * FROM [FactResellerSales] WHERE [CarrierTrackingNumber] = '4911-403C-9...'

Execution plan details:

- Nested Loops (Inner Join): Cost: 0 %
- Index Seek ([FactResellerSales].[CarrierTrackingNumber]): Cost: 6 %
- Key Lookup ([FactResellerSales].[SalesOrderNumber]): Cost: 94 %

Properties - Index Seek (NonClustered)

Property	Value
Actual Execution Mode	Row
Actual I/O Statistics	
Actual Number of Batches	0
Actual Number of Rows for All Batches	12
Actual Number of Rows Read	12
Actual Rebinds	0
Actual Rewinds	0
Actual Time Statistics	
Defined Values	[AdventureWorksDW2022].[dbo].[FactResellerSales].[CarrierTrackingNumber]
Description	Scan a particular range of rows from a nonclustered index
Estimated CPU Cost	0.000175
Estimated Execution Mode	Row
Estimated I/O Cost	0.003125
Estimated Number of Execution Plans	1
Estimated Number of Rows for All Batches	16,35
Estimated Number of Rows Per Batch	16,35
Estimated Number of Rows to be Examined	16,35
Estimated Operator Cost	0.0033 (6%)
Estimated Rebinds	0
Estimated Rewinds	0
Estimated Row Size	58 B
Estimated Subtree Cost	0.0033
Forced Index	False
ForceScan	False
ForceSeek	False
Logical Operation	Index Seek
Node ID	1
NoExpandHint	False
Number of Executions	1
Object	[AdventureWorksDW2022].[dbo].[FactResellerSales]
Ordered	True

Index to be used check -

SQLQuery2.sql - D:\BQBU\Youtube (52)* SQLQuery1.sql - D:\BQBU\Youtube (58)*

```
SELECT *
FROM FactResellerSales
WHERE CarrierTrackingNumber = '4911-403C-9...
```

CREATE NONCLUSTERED INDEX idx_FactReseller
ON FactResellerSales (CarrierTrackingNumber)

Query 1: Query cost (relative to the batch): 1

SELECT * FROM [FactResellerSales] WHERE [CarrierTrackingNumber] = '4911-403C-9...'

Execution plan details:

- Nested Loops (Inner Join): Cost: 0 %
- Index Seek ([FactResellerSales].[CarrierTrackingNumber]): Cost: 6 %
- Key Lookup ([FactResellerSales].[SalesOrderNumber]): Cost: 94 %

Properties - Index Seek (NonClustered)

Property	Value
Actual Rebinds	0
Actual Rewinds	0
Actual Time Statistics	
Defined Values	[AdventureWorksDW2022].[dbo].[FactResellerSales].[CarrierTrackingNumber]
Description	Scan a particular range of rows from a nonclustered index
Estimated CPU Cost	0.000175
Estimated Execution Mode	Row
Estimated I/O Cost	0.003125
Estimated Number of Execution Plans	1
Estimated Number of Rows for All Batches	16,35
Estimated Number of Rows Per Batch	16,35
Estimated Number of Rows to be Examined	16,35
Estimated Operator Cost	0.0033 (6%)
Estimated Rebinds	0
Estimated Rewinds	0
Estimated Row Size	58 B
Estimated Subtree Cost	0.0033
Forced Index	False
ForceScan	False
ForceSeek	False
Logical Operation	Index Seek
Node ID	1
NoExpandHint	False
Number of Executions	1
Object	[AdventureWorksDW2022].[dbo].[FactResellerSales]
Database	[AdventureWorksDW2022]
Index	[idx_FactReseller_CTA]
Index Kind	NonClustered
Schema	[dbo]
Storage	RowStore
Table	[FactResellerSales]
Ordered	True

SQLQuery2.sql - D:\8QBU\Youtube (52)* SQLQuery1.sql - D:\8QBU\Youtube (58)*

```

SELECT *
FROM FactResellerSales
WHERE CarrierTrackingNumber = '4911-403C-9'

CREATE NONCLUSTERED INDEX idx_FactResellerSales
ON FactResellerSales (CarrierTrackingNumber)

```

Query 1: Query cost (relative to the batch): 1.0000000000000000
 SELECT * FROM [FactResellerSales] WHERE [CarrierTrackingNumber] = '4911-403C-9'

Execution plan details:
 Nested Loops (Inner Join)
 Cost: 0.000s
 12 of 16 (75%)
 Index Seek ([FactResellerSales].[CarrierTrackingNumber])
 Cost: 6.000s
 12 of 16 (75%)
 Key Lookup ([FactResellerSales].[SalesOrderNumber])
 Cost: 94.000s
 12 of 16 (75%)

Properties: Index Seek (NonClustered)
 Actual Execution Mode: Row
 Actual I/O Statistics:
 Actual Number of Batches: 0
 Actual Number of Rows for All: 12
 Actual Number of Rows Read: 12
 Actual Rebinds: 0
 Actual Rewinds: 0
 Actual Time Statistics:
 Defined Values: [AdventureWorksDW2022].[FactResellerSales].[CarrierTrackingNumber]
 Description: Scan a particular range of rows
 Estimated CPU Cost: 0.000175
 Estimated Execution Mode: Row
 Estimated I/O Cost: 0.003125
 Estimated Number of Execution: 1
 Estimated Number of Rows for: 16,35
 Estimated Number of Rows Per: 16,35
 Estimated Number of Rows to be Scanned: 16,35
 Estimated Operator Cost: 0.0033 (6%)
 Estimated Rebinds: 0
 Estimated Rewinds: 0
 Estimated Row Size: 58 B
 Estimated Subtree Cost: 0.0033
 Forced Index: False
 ForceScan: False
 ForceSeek: False
 Logical Operation: Index Seek
 Node ID: 1
 NoExpandHint: False
 Number of Executions: 1
 Object: [AdventureWorksDW2022].[FactResellerSales].[CarrierTrackingNumber]
 Ordered: True

Now in table scan or heap execution

Here reading rows more than 60 thousand in order to get 12 number

SQLQuery2.sql - D:\8QBU\Youtube (52)* SQLQuery1.sql - D:\8QBU\Youtube (58)*

```

SELECT *
FROM FactResellerSales_HP
WHERE CarrierTrackingNumber = '4911-403C-9'

```

Query 1: Query cost (relative to the batch): 1.0000000000000000
 SELECT * FROM [FactResellerSales_HP] WHERE [CarrierTrackingNumber] = '4911-403C-9'

Execution plan details:
 Table Scan ([FactResellerSales_HP].[CarrierTrackingNumber])
 Cost: 100.000s
 12 of 15 (80%)

Properties: Table Scan
 Actual Execution Mode: Row
 Actual I/O Statistics:
 Actual Number of Batches: 0
 Actual Number of Rows for All: 12
 Actual Number of Rows Read: 60855
 Actual Rebinds: 0
 Actual Rewinds: 0
 Actual Time Statistics:
 Defined Values: [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[CarrierTrackingNumber]
 Description: Scan rows from a table.
 Estimated CPU Cost: 0.067019
 Estimated Execution Mode: Row
 Estimated I/O Cost: 1,23802
 Estimated Number of Execution: 1
 Estimated Number of Rows for: 14,6287
 Estimated Number of Rows Per: 14,6287
 Estimated Number of Rows to be Scanned: 14,6287
 Estimated Operator Cost: 1,30504 (100%)
 Estimated Rebinds: 0
 Estimated Rewinds: 0
 Estimated Row Size: 223 B
 Estimated Subtree Cost: 1,30504
 Forced Index: False
 ForceScan: False
 Logical Operation: Table Scan
 Node ID: 0
 NoExpandHint: False
 Number of Executions: 1
 Object: [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[CarrierTrackingNumber]
 Ordered: False
 Output List: [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[CarrierTrackingNumber]

Key look up plan -

Form key look up we get rest of all columns because we used * in code

The screenshot displays the SQL Server Enterprise Manager interface. The central pane shows a query window with the following SQL code:

```
SELECT *
FROM FactResellerSales
WHERE CarrierTrackingNumber = '4911-403C-98'

CREATE NONCLUSTERED INDEX idx_FactReseller_CTA
ON FactResellerSales (CarrierTrackingNumber)
```

The execution plan for the query is shown below the code. It consists of a 'SELECT' operator (cost 0%) and an 'Index Scan' operator (cost 6%). The 'Index Scan' operator is highlighted with a yellow box. The 'SELECT' operator shows 12 rows (75%) and 16 rows (75%). The 'Index Scan' operator shows 12 rows (75%) and 16 rows (75%). A 'Key Lookup' operator is also shown, which is highlighted with a yellow box. The 'Key Lookup' operator shows 12 rows (75%) and 16 rows (75%).

The right pane shows the 'Properties' window for the 'Key Lookup (Clustered)' operator. The 'Misc' tab is selected, showing various statistics and costs. The 'Actual Number of Rows Read' is 12. The 'Estimated CPU Cost' is 0.0001581. The 'Estimated Execution Mode' is 'Row'. The 'Estimated I/O Cost' is 0.003125. The 'Estimated Number of Execution' is 16,35. The 'Estimated Number of Rows for / 16,35' is 1. The 'Estimated Operator Cost' is 0.0503485 (94%). The 'Estimated Subtree Cost' is 0.0503485. The 'Forced Index' is 'False'. The 'ForceScan' is 'False'. The 'ForceSeek' is 'False'. The 'Logical Operation' is 'Key Lookup'. The 'Lookup' is 'True'. The 'Node ID' is 3. The 'NoExpandHint' is 'False'. The 'Number of Executions' is 12. The 'Object' is 'AdventureWorksDW2022'. The 'Ordered' is 'True'.

That's we we are able to read 12 rows

Like from index scan we got carrier tracking number and from key loop up we get all other values.

This operation is called nest loop

The screenshot displays the SQL Server Enterprise Manager interface. The central pane shows the same query as the previous screenshot:

```
SELECT *
FROM FactResellerSales
WHERE CarrierTrackingNumber = '4911-403C-98'

CREATE NONCLUSTERED INDEX idx_FactReseller_CTA
ON FactResellerSales (CarrierTrackingNumber)
```

The execution plan for the query is shown below the code. It consists of a 'SELECT' operator (cost 0%) and an 'Index Scan' operator (cost 6%). The 'Index Scan' operator is highlighted with a yellow box. The 'SELECT' operator shows 12 rows (75%) and 16 rows (75%). The 'Index Scan' operator shows 12 rows (75%) and 16 rows (75%). A 'Key Lookup' operator is also shown, which is highlighted with a yellow box. The 'Key Lookup' operator shows 12 rows (75%) and 16 rows (75%).

The right pane shows the 'Properties' window for the 'Nested Loops' operator. The 'Misc' tab is selected, showing various statistics and costs. The 'Actual Execution Mode' is 'Row'. The 'Actual Number of Batches' is 0. The 'Actual Number of Rows for All E 12' is 0. The 'Actual Rebinds' is 0. The 'Actual Rewinds' is 0. The 'Actual Time Statistics' are shown for each row in the top (0). The 'Description' is 'For each row in the top (0)'. The 'Estimated CPU Cost' is 0.0000683. The 'Estimated Execution Mode' is 'Row'. The 'Estimated I/O Cost' is 0. The 'Estimated Number of Execution' is 1. The 'Estimated Number of Rows for / 16,35' is 16,35. The 'Estimated Operator Cost' is 0.0000683 (0%). The 'Estimated Subtree Cost' is 0.0537168. The 'Forced Index' is 'False'. The 'ForceScan' is 'False'. The 'ForceSeek' is 'False'. The 'Logical Operation' is 'Nested Loops'. The 'Node ID' is 0. The 'Number of Executions' is 1. The 'Optimized' is 'False'. The 'Outer References' are 'AdventureWorksDW2022'. The 'Output List' is 'AdventureWorksDW2022'. The 'Parallel' is 'False'. The 'Physical Operation' is 'Nested Loops'.



Join Algorithms



Nested Loops: Compares tables row by row; best for small tables.



Hash Match: Matches rows using a hash table; best for large tables.



Merge Join: Merge two sorted tables; efficient when both are sorted.

For large tables hash match and merge join is very good when you have
When you have large amount of data from index and key look ups and sql using
nested loop it's not good.

Query 1: Query cost (relative to the batch): 100%

SELECT * FROM [FactResellerSales] WHERE [CarrierTrackingNum...

Execution plan details:

- Nested Loops (Inner Join) Cost: 0 %
- Index Seek on [FactResellerSales] Cost: 6 %
- Key Lookup on [FactResellerSales] Cost: 94 %

Properties window details for Index Seek (NonClustered):

Misc	Row
Actual Execution Mode	Row
Actual I/O Statistics	
Actual Number of Batches	0
Actual Number of Rows for All	12
Actual Number of Rows Read	12
Actual Rebinds	0
Actual Rewinds	0
Actual Time Statistics	
Defined Values	[AdventureWorksDW2022]
Description	Scan a particular range of
Estimated CPU Cost	0,000175
Estimated Execution Mode	Row
Estimated I/O Cost	0,003125
Estimated Number of Execution	1
Estimated Number of Rows for	16,35
Estimated Number of Rows to b	16,35
Estimated Operator Cost	0,0033 (6%)
Estimated Rebinds	0
Estimated Rewinds	0
Estimated Row Size	58 B
Estimated Subtree Cost	0,0033
Forced Index	False
ForceScan	False
ForceSeek	False
Logical Operation	Index Seek
Node ID	1
NoExpandHint	False
Number of Executions	1
Object	[AdventureWorksDW2022]
Ordered	True

Heap structure cost

The screenshot shows a SQL query in the query editor:

```
SELECT *
FROM FactResellerSales_HP
WHERE CarrierTrackingNumber = '4911-403C-98'
```

The execution plan for Query 1 is displayed below the query. The query cost (relative to the batch) is 100%. The plan shows a single step: a Table Scan on [FactResellerSales_HP] with a cost of 100.000s, 12 of 15 rows returned (80%).

No look up needed if you remove * from query it will search for specific carrier number from index only

The screenshot shows a SQL query in the query editor:

```
SELECT CarrierTrackingNumber
FROM FactResellerSales
WHERE CarrierTrackingNumber = '4911-403C-98'
```

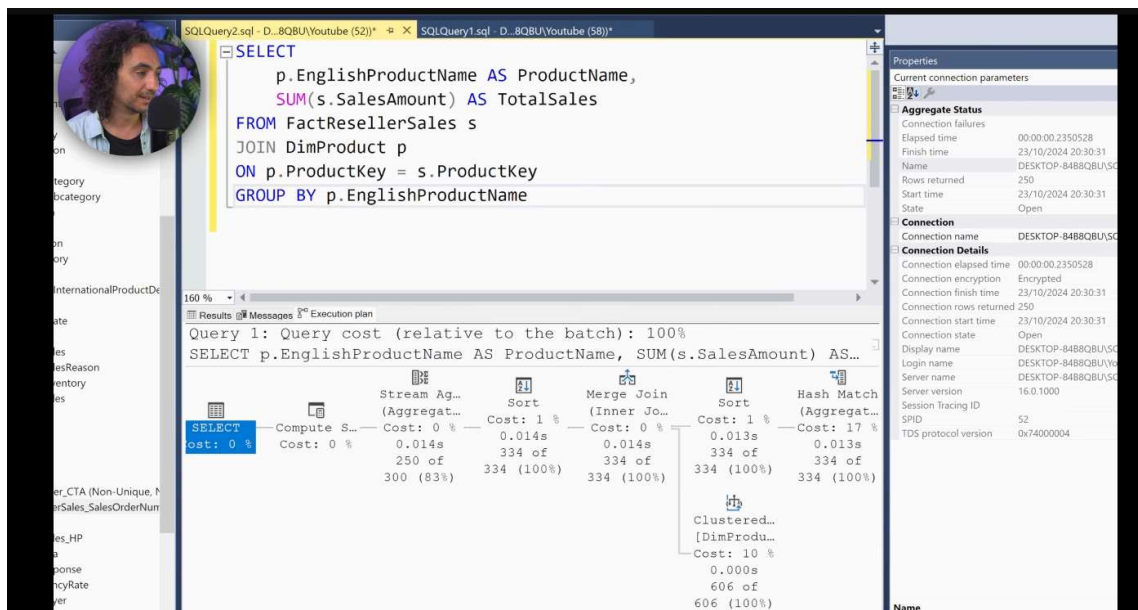
The execution plan for Query 1 is displayed below the query. The query cost (relative to the batch) is 100%. The plan shows a single step: an Index Seek on [FactResellerSales] with a cost of 100.000s, 12 of 16 rows returned (75%).

On the right side of the screenshot, the Properties window is open, showing the current connection parameters:

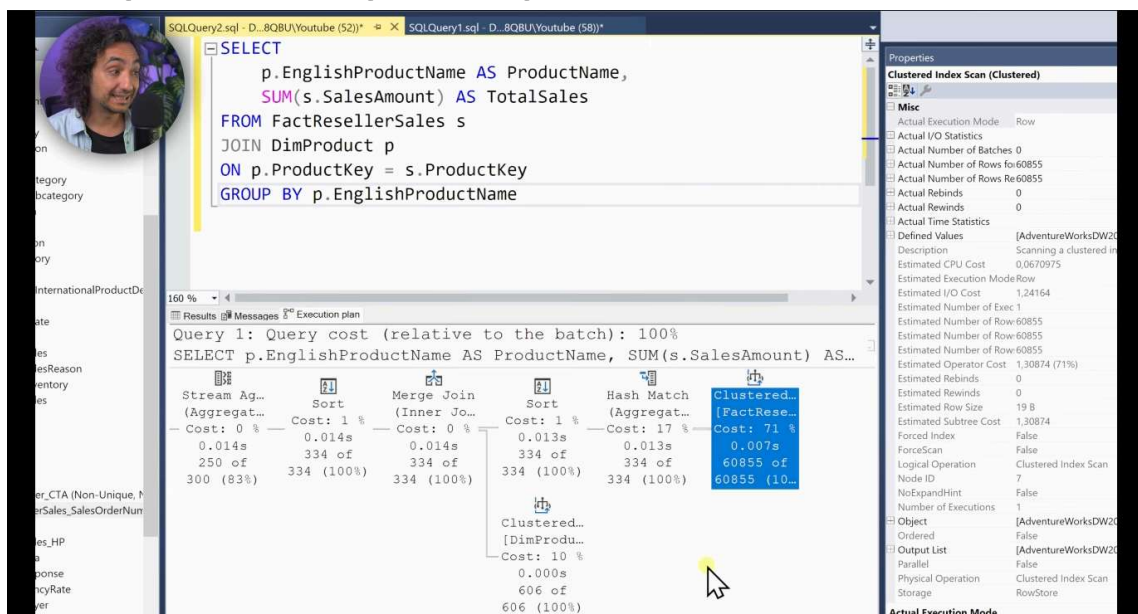
Current connection parameters	
Aggregate Status	
Connection failures	
Elapsed time	00:00:00.1526081
Finish time	23/10/2024 20:07:13
Name	DESKTOP-84B8Q8U\SQLE
Rows returned	12
Start time	23/10/2024 20:07:12
State	Open
Connection	
Connection name	DESKTOP-84B8Q8U\SQLE
Connection Details	
Connection elapsed time	00:00:00.1526081
Connection encryption	Encrypted
Connection finish time	23/10/2024 20:07:13
Connection rows returned	12
Connection start time	23/10/2024 20:07:12
Connection state	Open
Display name	DESKTOP-84B8Q8U\SQLE
Login name	DESKTOP-84B8Q8U\SQLE
Server name	DESKTOP-84B8Q8U\SQLE
Server version	16.0.1000
Session Tracing ID	
SPID	52
TDS protocol version	0x74000004

Execution Plan

Rowstore vs. Columnstore



Checking from left two right. Getting data from fast



Hash match from aggregation

Sorting data because it's doing later inner join

Next clustered scan is for dimensions

It's around 606 rows for clustered scan and result of it is sorted

SQLQuery2.sql - D:\8QBU\Youtube (52)* X SQLQuery1.sql - D:\8QBU\Youtube (58)*

```
SELECT
    p.EnglishProductName AS ProductName,
    SUM(s.SalesAmount) AS TotalSales
FROM FactResellerSales s
JOIN DimProduct p
ON p.ProductKey = s.ProductKey
GROUP BY p.EnglishProductName
```

Query 1: Query cost (relative to the batch): 100%

SELECT p.EnglishProductName AS ProductName, SUM(s.SalesAmount) AS...

Stream Agg... (Aggregat... Cost: 0 % 0.014s 250 of 300 (83%))

Sort (Inner Jo... Cost: 1 % 0.014s 334 of 334 (100%))

Merge Join (Inner Jo... Cost: 0 % 0.014s 334 of 334 (100%))

Sort (Inner Jo... Cost: 1 % 0.013s 334 of 334 (100%))

Hash Match (Aggregat... [FactRese... Cost: 17 % 0.013s 334 of 334 (100%))

Clustered... (DimProdu... Cost: 71 % 0.007s 60855 of 60855 (100%))

Clustered... [DimProdu... Cost: 10 % 0.000s 606 of 606 (100%)

Properties

Clustered Index Scan (Clustered)

Misc

- Actual Execution Mode: Row
- Actual I/O Statistics
- Actual Number of Batches: 0
- Actual Number of Rows for Batch: 606
- Actual Number of Rows for Batch: 606
- Actual Rebinds: 0
- Actual Rewinds: 0
- Actual Time Statistics
- Defined Values: [AdventureWorksDW2008]
- Description: Scanning a clustered index
- Estimated CPU Cost: 0.0008236
- Estimated Execution Mode: Row
- Estimated I/O Cost: 0.18831
- Estimated Number of Exec: 1
- Estimated Number of Rows: 606
- Estimated Number of Rows for Batch: 606
- Estimated Operator Cost: 0.189134 (10%)
- Estimated Rebinds: 0
- Estimated Rewinds: 0
- Estimated Row Size: 53 B
- Estimated Subtree Cost: 0.189134
- Forced Index: False
- ForceScan: False
- ForceSeek: False
- Logical Operation: Clustered Index Scan
- Node ID: 8
- NoExpandHint: False
- Number of Executions: 1
- Object: [AdventureWorksDW2008].[FactResellerSales]
- Ordered: True
- Output List: [AdventureWorksDW2008].[FactResellerSales].[EnglishProductName], [AdventureWorksDW2008].[FactResellerSales].[SalesAmount]
- Parallel: False
- Physical Operation: Clustered Index Scan

Actual Execution Mode

Together two sorted data now so next operation is merge join to join two sorted data set

Then data got sorted and % it as output then return output

First step clustered - cost is high

SQLQuery2.sql - D:\8QBU\Youtube (52)* SQLQuery1.sql - D:\8QBU\Youtube (58)*

```

SELECT
    p.EnglishProductName AS ProductName,
    SUM(s.SalesAmount) AS TotalSales
FROM FactResellerSales s
JOIN DimProduct p
ON p.ProductKey = s.ProductKey
GROUP BY p.EnglishProductName

```

Query 1: Query cost (relative to the batch): 100%

Stream Ag... (Aggregat... Cost: 0 % 0.014s 250 of 300 (83%))

Sort (Inner Jo... Cost: 1 % 0.014s 334 of 334 (100%))

Merge Join (Inner Jo... Cost: 0 % 0.013s 334 of 334 (100%))

Hash Match (Aggregat... Cost: 17 % 0.013s 334 of 334 (100%))

Clustered... [FactRese... Cost: 71 % 0.007s 60855 of 60855 (100%)]

Properties

Clustered Index Scan (Clustered)

Misc

Actual Execution Mode Row

Actual I/O Statistics

Actual Number of Batches 0

Actual Number of Rows for 60855

Actual Number of Rows for 60855

Actual Rebinds 0

Actual Rewinds 0

Actual Time Statistics

Defined Values [AdventureWorksDW2019]

Description Scanning a clustered index

Estimated CPU Cost 0.0670975

Estimated Execution Mode Row

Estimated I/O Cost 1.24164

Estimated Number of Exec 1

Estimated Number of Rows 60855

Estimated Number of Rows 60855

Estimated Operator Cost 1.30874 (71%)

Estimated Rebinds 0

Estimated Rewinds 0

Estimated Row Size 19 B

Estimated Subtree Cost 1.30874

Forced Index False

ForceScan False

Logical Operation Clustered Index Scan

Node ID 7

NoExpandHint False

Number of Executions 1

Object [AdventureWorksDW2019]

Ordered False

Output List [AdventureWorksDW2019]

Parallel False

Physical Operation Clustered Index Scan

Storage RowStore

Actual Execution Mode

Above code is heap structure now below we are using index in below code
To optimize this we can use column store index

If you are doing aggregation on big table then column store index is good idea

```

SELECT
    p.EnglishProductName AS ProductName,
    SUM(s.SalesAmount) AS TotalSales
FROM FactResellerSales_HP s
JOIN DimProduct p
ON p.ProductKey = s.ProductKey
GROUP BY p.EnglishProductName

```

Query 1: Query cost (relative to the batch): 100%

SELECT p.EnglishProductName AS ProductName, SUM(s.SalesAmount) AS...

Compute S... (Aggregat... Cost: 0 % 0.000s 250 of 300 (83%))

Hash Match (Aggregat... Cost: 2 % 0.000s 250 of 300 (83%))

Hash Match (Inner Jo... Cost: 1 % 0.000s 334 of 334 (100%))

Hash Match (Aggregat... Cost: 13 % 0.001s 334 of 334 (100%))

Columnsto... [FactRese... Cost: 6 % 0.000s 60855 of 60855 (100%)]

Filter (DimProdu... Cost: 79 % 0.000s 334 of 606 of 606 (100%))

Properties

Current connection parameters

Aggregate Status

Connection failures

Elapsed time 00:00:00.1806005

Finish time 23/10/2024 20:38:05

Name DESKTOP-8488QBU\SC

Rows returned 250

Start time 23/10/2024 20:38:05

State Open

Connection

Connection name DESKTOP-8488QBU\SC

Connection Details

Connection elapsed time 00:00:00.1806005

Connection encryption Encrypted

Connection finish time 23/10/2024 20:38:05

Connection rows returned 250

Connection start time 23/10/2024 20:38:05

Connection state Open

Display name DESKTOP-8488QBU\SC

Login name DESKTOP-8488QBU\SC

Server name DESKTOP-8488QBU\SC

Server version 16.0.1000

Session Tracing ID

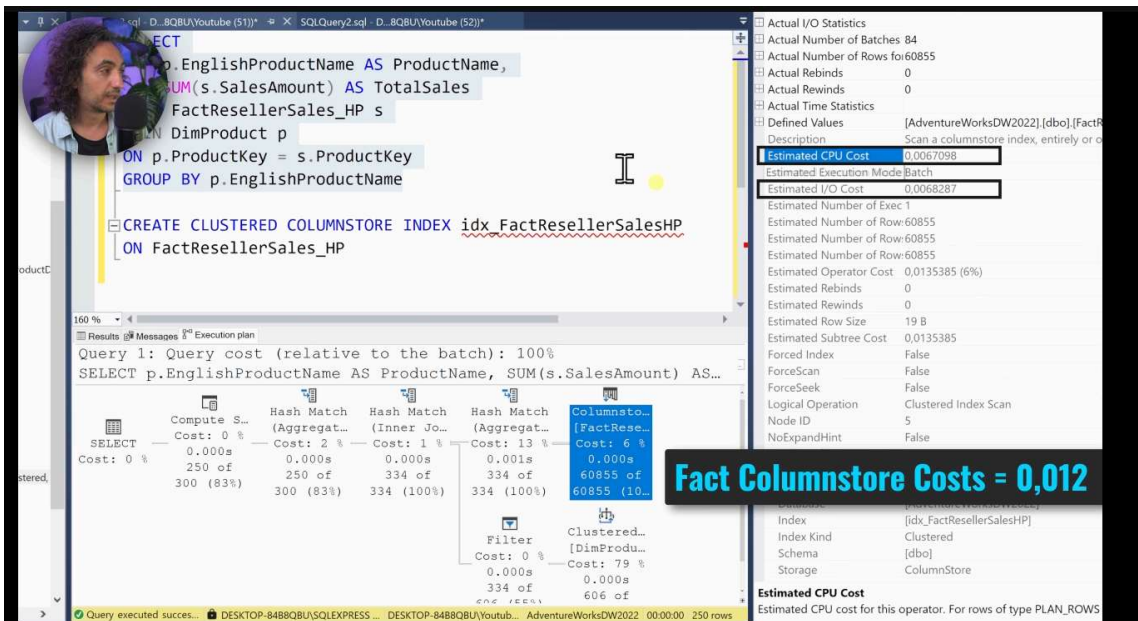
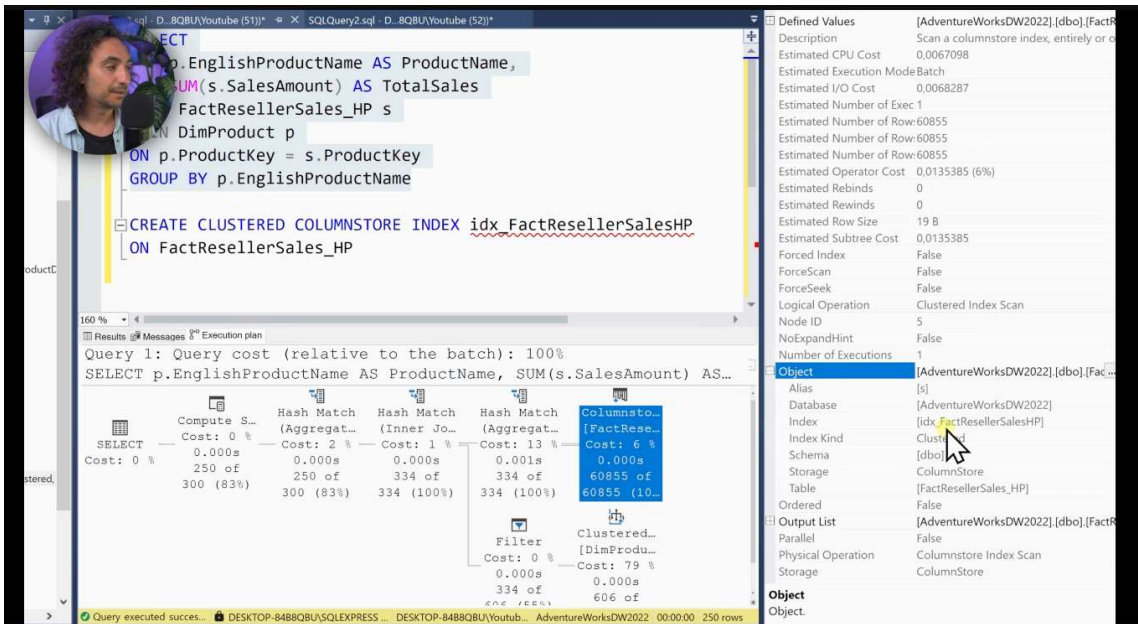
SPID 51

TDS protocol version 0x74000004

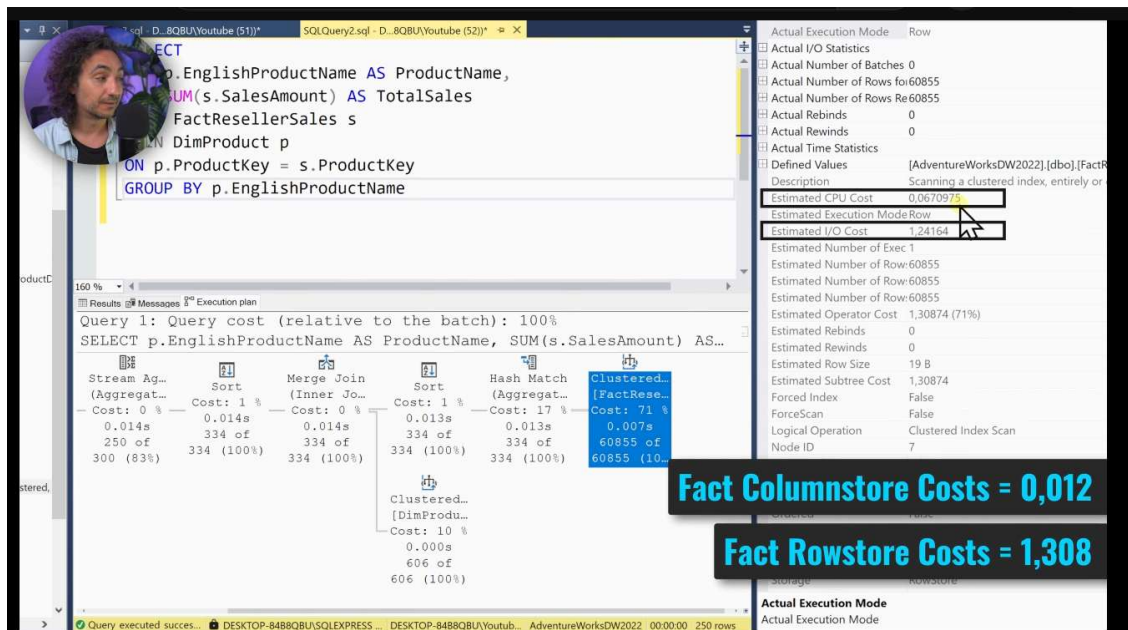
Name

Now cost is 6%

Whole data store only index



Below is cost for heap structure



Query 1: Query cost (relative to the batch): 100%

```
SELECT p.EnglishProductName AS ProductName, SUM(s.SalesAmount) AS...
```

Stream Agg... (Aggregat... Cost: 0 % 0.014s 250 of 300 (83%))

Sort (Aggregat... Cost: 1 % 0.014s 334 of 334 (100%))

Merge Join (Inner Jo... Cost: 0 % 0.014s 334 of 334 (100%))

Sort (Aggregat... Cost: 1 % 0.013s 334 of 334 (100%))

Hash Match (Aggregat... Cost: 17 % 0.013s 334 of 334 (100%))

Clustered... [DimProdu... Cost: 10 % 0.000s 606 of 606 (100%)

Actual Execution Mode Row

Actual I/O Statistics

Actual Number of Batches 0

Actual Number of Rows for 60855

Actual Number of Rows Re 60855

Actual Rebinds 0

Actual Rewinds 0

Actual Time Statistics

Defined Values [AdventureWorksDW2022].[dbo].[FactR...

Description Scanning a clustered index, entirely or

Estimated CPU Cost 0.0670975

Estimated Execution Mode Row

Estimated I/O Cost 1.24164

Estimated Number of Exec 1

Estimated Number of Row 60855

Estimated Number of Row 60855

Estimated Number of Row 60855

Estimated Operator Cost 1.30874 (71%)

Estimated Rebinds 0

Estimated Rewinds 0

Estimated Row Size 19 B

Estimated Subtree Cost 1.30874

Forced Index False

ForceScan False

Logical Operation Clustered Index Scan

Node ID 7

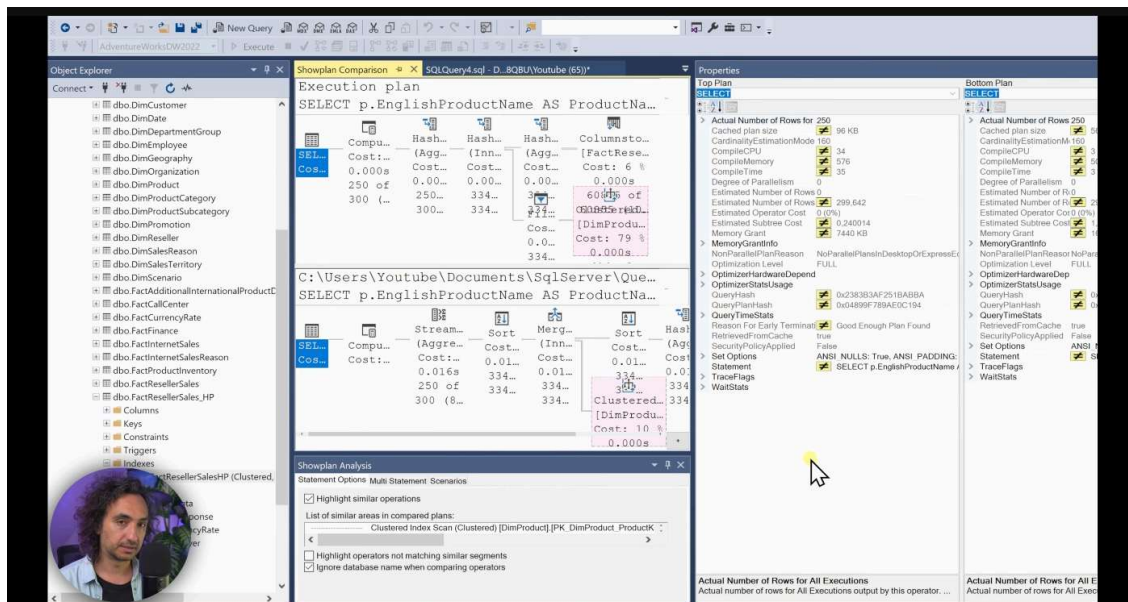
Fact Columnstore Costs = 0,012

Fact Rowstore Costs = 1,308

Actual Execution Mode

Actual Execution Mode

You can right click on execution plan and save execution plan then go again on second one go right click choose compare and show show execution plan and get more details.



Object Explorer

Connect

AdventureWorksDW2022

SQLQuery4.sql - D:\BQBU\Youtube (65)*

Execution plan

```
SELECT p.EnglishProductName AS ProductNa...
```

Compu... Hash... Hash... Hash... Columnsto... Cost: 0.000s 250 of 300 (83%)

Sort (Aggregat... Cost: 1 % 0.014s 334 of 334 (100%))

Merge Join (Inner Jo... Cost: 0 % 0.014s 334 of 334 (100%))

Sort (Aggregat... Cost: 1 % 0.013s 334 of 334 (100%))

Hash Match (Aggregat... Cost: 17 % 0.013s 334 of 334 (100%))

Clustered... [DimProdu... Cost: 10 % 0.000s 606 of 606 (100%)

C:\Users\Youtube\Documents\SqlServer\Que...

SELECT p.EnglishProductName AS ProductNa...

Compu... Stream... Sort... Merge... Hash... Cost: 0.000s 250 of 300 (83%)

Sort (Aggregat... Cost: 1 % 0.014s 334 of 334 (100%))

Merge Join (Inner Jo... Cost: 0 % 0.014s 334 of 334 (100%))

Sort (Aggregat... Cost: 1 % 0.013s 334 of 334 (100%))

Hash Match (Aggregat... Cost: 17 % 0.013s 334 of 334 (100%))

Clustered... [DimProdu... Cost: 10 % 0.000s 606 of 606 (100%)

Showplan Analysis

Statement Options Multi Statement Scenarios

Highlight similar operations

List of similar areas in compared plans:

Clustered Index Scan (Clustered) [DimProduct] [PK_DimProduct_ProductKey]

Highlight operators not matching similar segments

Ignore database name when comparing operators

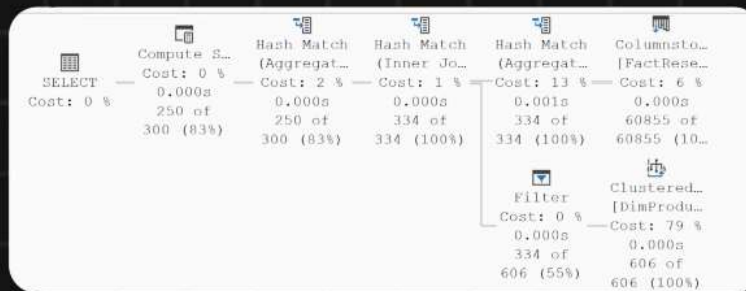
Actual Number of Rows for All Executions

Actual number of rows for All Executions output by this operator...

Actual Number of Rows for All Executions

Actual number of rows for All Executions output by this operator...

Execution Plan



Understand how SQL executes your query

How many resources your query consumes?

Check if your new Indexes are used

Testing & Experimenting Indexes

SQL HINTS

Bad Execution Plan!

1. Outdated Statistics

2. Too Many Indexes

Here we need sql hints

SQL HINTS

Commands you add to a query to force the database to run it in a specific way for better performance

Explorer

ESKTOP-84B8QBU\SQLEXPRESS (SQL Server)

- Databases
 - System Databases
 - Database Snapshots
 - AdventureWorks2022
 - AdventureWorksDW2022
 - SalesDB
 - Database Diagrams
 - Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - Sales.Customers
 - Columns
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - idx_Customers_Country (Non-Clustered)
 - idx_Customers_CS_Country (Non-Clustered)
 - PK_Customer_A4AE64B87FC
 - Statistics
 - Sales.CustomersTest
 - Sales.DBCustomers
 - Sales.Employees
 - Sales.Orders
 - Sales.OrdersArchive
 - Sales.Products
 - Views
 - External Resources
 - Synonyms

SQLQuery1.sql D:\8QBU\Youtube (72)*

```
SELECT
    o.Sales,
    c.Country
FROM Sales.Orders o
LEFT JOIN Sales.Customers c
ON
    o.CustomerID = c.CustomerID
```

159 %

Query 1: Query cost (relative to the batch): 100%

SELECT o.Sales, c.Country FROM Sales.Orders o LEFT JOIN Sales.Customers c

Execution plan

Nested Loops (Left Out...)

Cost: 3 %

0.000s

10 of

10 (100%)

Clustered Index [Orders]...

Cost: 44 %

0.000s

10 of

10 (100%)

Clustered Index [Customers]...

Cost: 53 %

0.000s

50 of

50 (100%)

Explorer

ESKTOP-84B8QBU\SQLEXPRESS (SQL Server)

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 - External Resources
 - Synonyms

SQLQuery1.sql D:\8QBU\Youtube (72)*

```
SELECT
    o.Sales,
    c.Country
FROM Sales.Orders o
LEFT JOIN Sales.Customers c
ON
    o.CustomerID = c.CustomerID
OPTION (HASH JOIN)
```

159 %

Query 1: Query cost (relative to the batch): 100%

SELECT o.Sales, c.Country FROM Sales.Orders o LEFT JOIN Sales.Customers c

Execution plan

Hash Match (Left Out...)

Cost: 22 %

0.000s

10 of

10 (100%)

Clustered Index [Orders]...

Cost: 40 %

0.000s

10 of

10 (100%)

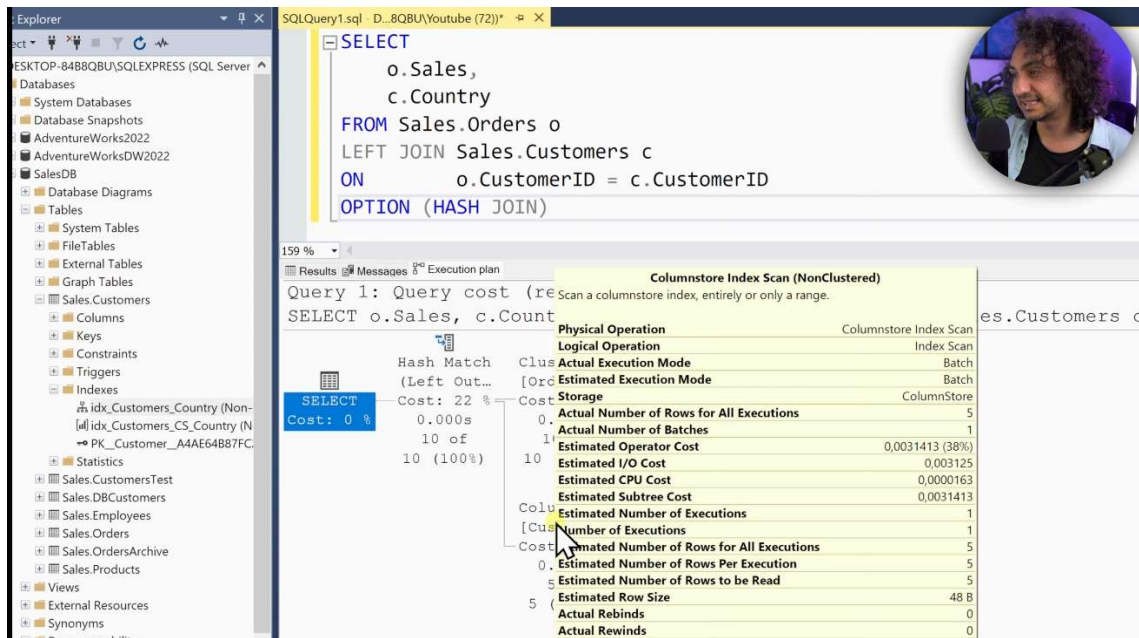
Columnstore Index [Customers]...

Cost: 38 %

0.009s

5 of

5 (100%)



SQLQuery1.sql - D:\8QBU\Youtube (72)*

```

SELECT
    o.Sales,
    c.Country
FROM Sales.Orders o
LEFT JOIN Sales.Customers c
ON
    o.CustomerID = c.CustomerID
OPTION (HASH JOIN)

```

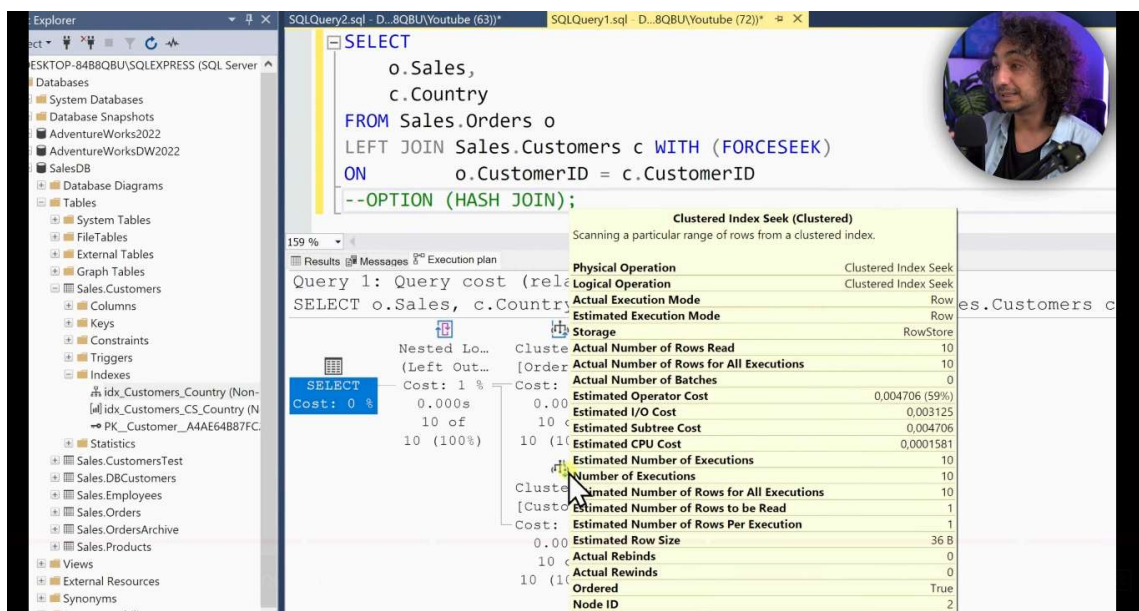
Query 1: Query cost (relational algebra)

Columnstore Index Scan (NonClustered)

Scan a columnstore index, entirely or only a range.

Physical Operation	Columnstore Index Scan
Logical Operation	Index Scan
Actual Execution Mode	Batch
Estimated Execution Mode	Batch
Storage	ColumnStore
Actual Number of Rows for All Executions	5
Actual Number of Batches	1
Estimated Operator Cost	0.0031413 (38%)
Estimated I/O Cost	0.003125
Estimated CPU Cost	0.0000163
Estimated Subtree Cost	0.0031413
Estimated Number of Executions	1
Number of Executions	1
Estimated Number of Rows for All Executions	5
Estimated Number of Rows Per Execution	5
Estimated Number of Rows to be Read	5
Estimated Row Size	48 B
Actual Rebinds	0
Actual Rewinds	0

Changed index seek to index scan



SQLQuery2.sql - D:\8QBU\Youtube (63)*

```

SELECT
    o.Sales,
    c.Country
FROM Sales.Orders o
LEFT JOIN Sales.Customers c WITH (FORCESEEK)
ON
    o.CustomerID = c.CustomerID
--OPTION (HASH JOIN);

```

Query 1: Query cost (relational algebra)

Clustered Index Seek (Clustered)

Scanning a particular range of rows from a clustered index.

Physical Operation	Clustered Index Seek
Logical Operation	Clustered Index Seek
Actual Execution Mode	Row
Estimated Execution Mode	Row
Storage	RowStore
Actual Number of Rows Read	10
Actual Number of Rows for All Executions	10
Actual Number of Batches	0
Estimated Operator Cost	0.004706 (59%)
Estimated I/O Cost	0.003125
Estimated Subtree Cost	0.004706
Estimated CPU Cost	0.0001581
Estimated Number of Executions	10
Number of Executions	10
Estimated Number of Rows for All Executions	10
Estimated Number of Rows to be Read	1
Estimated Number of Rows Per Execution	1
Estimated Row Size	36 B
Actual Rebinds	0
Actual Rewinds	0
Ordered	True
Node ID	2

Hint to sql to use better index

SQLQuery2.sql - D:\8QBU\Youtube (63))

```

SELECT
    o.Sales,
    c.Country
FROM Sales.Orders o
LEFT JOIN Sales.Customers c WITH (INDEX([PK_Customer_A4AE64B87FC20A48]))
ON
    o.CustomerID = c.CustomerID
--OPTION (HASH JOIN);

```

Query 1: Query cost (relative to the batch): 100%

SELECT o.Sales, c.Country FROM Sales.Orders o LEFT JOIN Sales.Customers c WITH

Execution plan details:

- Nested Lo... (Left Out... [Orders]... Cost: 1 %
- Clustered... [Customers]... Cost: 41 %
- Cost: 0 %
- 0.000s
- 10 of
- 10 (100%)
- 10 (100%)
- 10 (100%)

Execution plan details:

- Actual Execution Mode
- Estimated Execution Mode
- Storage
- Actual Number of Rows Read
- Actual Number of Rows for All Executions
- Actual Number of Batches
- Estimated Operator Cost
- Estimated I/O Cost
- Estimated Subtree Cost
- Estimated CPU Cost
- Estimated Number of Executions
- Number of Executions
- Estimated Number of Rows for All Executions
- Estimated Number of Rows to be Read
- Estimated Number of Rows Per Execution
- Estimated Row Size
- Actual Rebinds
- Actual Rewinds
- Ordered
- Node ID
- Object
- Output List
- Seek Predicates

Query executed successfully.

These in development database these hints used good but in production db these not work may better

Because prod db is very large so you have test hint in each db to get correct optimization.

TIPS | SQL HINTS

1. Test hints in all project environments (DEV, PROD) as performance may vary.
2. Hints are quick fixes (Workaround not Solution)
You still have to find the cause and fix it.

Do not depend always on hints for permanent fix.